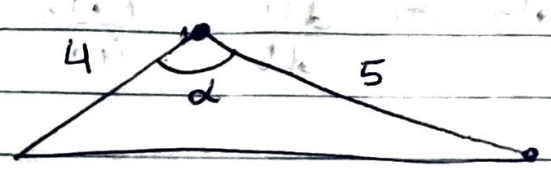
Lista 2

(2)



$$\frac{d\alpha}{dt} = 0{,}06$$

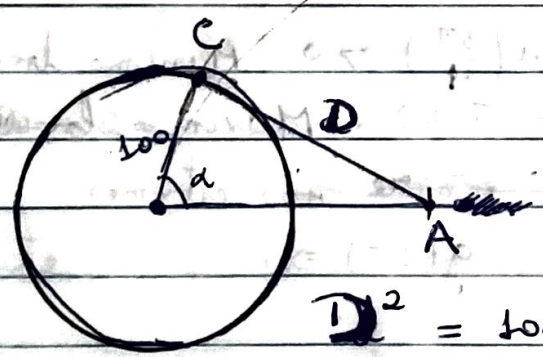
Fórmula da área em funcão do seno de α:

$$A = \frac{1}{2} a.b \operatorname{sen}\alpha = \frac{1}{2}.4.5 \operatorname{sen}\alpha$$

$$A = 10 \operatorname{sen}\alpha \Rightarrow \frac{dA}{dt} = 10.\cos\alpha.\frac{d\alpha}{dt}$$

quando $\alpha = \frac{\pi}{3}$, $\frac{dA}{dt} = 10.\frac{1}{2}.0{,}06 = 0{,}3$

(3)



$$D^2 = 100^2 + 200^2 - 2.100.200.\cos\alpha$$

$$2D.\frac{dD}{dt} = -400.(-\operatorname{sen}\alpha)\frac{d\alpha}{dt}$$

$$\alpha = \frac{l}{100}$$

$$\frac{d\alpha}{dt} = \frac{dl}{dt}.\frac{1}{100} = \frac{7}{100}$$

$$200^2 = h^2 + 50^2 \Rightarrow h^2 = 40.000 - 2500$$

$$= 37500 \Rightarrow h = 50\sqrt{15}$$

$$\operatorname{sen}\alpha = \frac{50\sqrt{15}}{200} = \frac{\sqrt{15}}{4}.$$

$$2 \cdot D \frac{dD}{dt} = 400 \cdot \frac{\sqrt{15}}{4} \cdot \frac{7}{100}$$

$$400 \frac{dD}{dt} = 7\sqrt{15} \Rightarrow \frac{dD}{dt} = \frac{7\sqrt{15}}{400}.$$

(8) $h(x) = |\operatorname{sen} x - \cos x|$

$$h(x)^2 = (\operatorname{sen} x - \cos x)^2 = \operatorname{sen}^2 x - 2 \operatorname{sen} x \cos x + \cos^2 x$$
$$= 1 - 2 \operatorname{sen} x \cos x = 1 - \operatorname{sen}(2x).$$

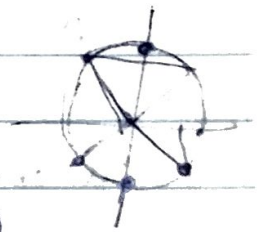
$g(x) = h(x)^2$.

$g'(x) = 0 \Rightarrow -\cos(2x) \cdot 2 = 0 \Rightarrow \cos(2x) = 0$

Em $[0, 2\pi]$, $2x = \frac{\pi}{2} + 2n\pi$ ou $2x = \frac{3\pi}{2} + 2n\pi$

$x = \frac{\pi}{4} + n\pi, \frac{3\pi}{4} + n\pi \Rightarrow \frac{\pi}{4}, \frac{5\pi}{4}$

$\frac{3\pi}{4}, \frac{7\pi}{4}$



$g''(x) = 4 \operatorname{sen}(2x)$.

$g''(\frac{\pi}{4}) = 4 \cdot \frac{\sqrt{2}}{2} = 2\sqrt{2} > 0$. Mínimo local

$g''(\frac{3\pi}{4}) = 4 \operatorname{sen}(\frac{3\pi}{4}) > 0$. Mínimo local

$g''(5\pi/4) = -2\sqrt{2} < 0$. Máximo local, idem $\frac{7\pi}{4}$

~~Logo, máximo ocorre~~ nos extremos.

$g(0) = 1$, $g(2\pi) = 1$.

Logo, máximo de $g(x)$ ocorre em $\frac{7\pi}{4}$.

Mesmo para $h(x)$:

$$\max h = \left| \operatorname{sen} \frac{5\pi}{4} - \cos \frac{5\pi}{4} \right| = \left| -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \right|$$
$$= \sqrt{2}.$$

(5) $a, b > 0$

$f(x) = x^a (1-x)^b \qquad 0 \leq x \leq 1$

$f(0) = f(1) = 0.$

$f'(x) = a x^{a-1} (1-x)^b + b x^a \cdot (1-x)^{b-1} \cdot (-1)$

$= a x^{a-1} (1-x)^b - b x^a (1-x)^{b-1}$

$f'(x) = 0 : \quad x^{a-1} (1-x)^{b-1} [a(1-x) - b \cdot x] = 0$

$-(a+b)x + a = 0 \Rightarrow x = \frac{a}{a+b}$ unico p. crítico

$f'(x) > 0 \Leftrightarrow \underbrace{x^{a-1} (1-x)^{b-1}}_{>0} [-(a+b)x + a] > 0$

(pois $x > 0$ e $1 - x > 0$, para $x \in (0, 1)$.)

$\Leftrightarrow -(a+b)x + a > 0 \Leftrightarrow x < \frac{a}{a+b}$

Analogamente, $f'(x) < 0$ sse $x > \frac{a}{a+b}$

Logo, $x = \frac{a}{a+b}$ é o ponto de máximo global em $[0, 1]$.

$f\left(\frac{a}{a+b}\right) = \frac{a^a}{(a+b)^a} \cdot \left(\frac{a+b}{a+b} - \frac{a}{a+b}\right)^b = \frac{a^a \cdot b^b}{(a+b)^{a+b}}.$

(9) $x^4 + 4x + c = 0$

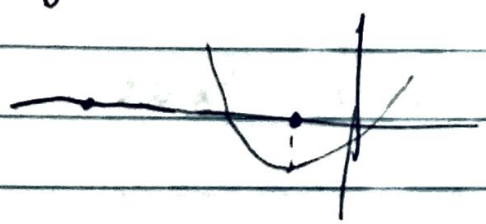
$f(x) = x^4 + 4x + c$

$f'(x) = 4x^3 + 4 > 0 \iff x^3 > -1 \iff x > -1.$

$f'(x) = 0 \iff x = -1.$

$f'(x) < 0 \iff x < -1.$

Logo, $x = -1$ é o ponto de mínimo global e único extremo local.



Se $f(-1) < 0$, como $f(-\infty) > 0$, existe um $x_0 < -1$, $f(x_0) = 0$ pelo TVI. Este é único em $(-\infty, -1)$. Do contrário se existisse $x_1 < -1$, $x_1 \neq x_0$; $f(x_1) = 0$, pelo TVM, existiria $x_2 \in (x_1, x_0)$ com $f'(x_2) = 0$, impossível. Por um argumento igual, há no máximo uma raiz de $f(x) = 0$ maior que $x_0 = -1$. Logo, $f(x) = 0$ tem no máximo 2 raízes.

(12) $f(x) = a x e^{bx^2}$

$f(2) = 1$: $a \cdot 2 e^{b \cdot 2^2} = 1 \Rightarrow 2 a e^{4b} = 1$

$\Rightarrow a = \dfrac{1}{2e^{4b}}$.

$$f(x) = \frac{x e^{bx^2}}{2e^{4b}}$$

$$f'(x) = \frac{e^{bx^2} + x \cdot e^{bx^2} \cdot 2xb}{2e^{4b}}$$

$f'(x) = 0$: $(1 + 2bx^2) e^{bx^2} = 0 \Rightarrow 1 + 2bx^2 = 0 \Rightarrow$

$2bx^2 = -1 \Rightarrow x^2 = \dfrac{-1}{2b} \Rightarrow b < 0, \; x = \pm\sqrt{\dfrac{-1}{2b}}$

$$f''(x) = \frac{1}{2e^{4b}} \cdot \left[4bx e^{bx^2} + (1 + 2bx^2) \cdot e^{bx^2} \cdot 2bx \right]$$

Se tem máximo em $x = 2$,

$$4 = \frac{-1}{2b} \Rightarrow b = \frac{-1}{8}$$

$$a = \frac{1}{2e^{-1/2}} = \frac{\sqrt{e}}{2}.$$

Obs: Não foi necessária $f''(x)$ acima.